

Persephonē

The Invisible Canals of Milan • Final Report

A Multimodal WebAR & Conversational-AI Portal to Milan's Buried Navigli

Project Links

- Live prototype** <https://persephone-orpin.vercel.app/> — open on mobile, grant camera / mic / location. AR and Photo Game require on-site Milan; map discovery work anywhere.
- GitHub repo** <https://github.com/Lorenzo-Chiaretti/Persephone>
- Demo video** [Click here to open the video](#)

1 Project Concept

Summary

Persephonē is a mobile-first web app combining **WebAR** (8th Wall / A-Frame) and **Conversational AI** to reveal Milan's buried Navigli canals. It overlays real-time 3D water reconstructions onto modern streets and lets users speak with the **Sciura di Milano**, a context-aware voice avatar — turning buried heritage into an embodied, emotional encounter.

Milan's Navigli were progressively buried under asphalt during the 20th century. Named from **Persephone**, who moves between surface and underworld, the project frames the experience as *revealing what is already there, just unseen*, answering a strong civic desire (**94%** in favour in Milan's 2011 referendum). Four design pillars drive the build:

- **Zero-App Access:** Nuxt 3 + 8th Wall + A-Frame launched from the browser via QR — no install friction.
- **Multimodal WebAR:** 8th Wall SLAM and custom water shaders anchor 3D reconstructions beneath modern street level.
- **Conversational AI:** Hands-free dialogue with the Sciura via Deepgram STT, Groq Llama 3.1, and ElevenLabs TTS.
- **Gamified Exploration:** GPS “hot-and-cold” photo hunt with acoustic Geiger-counter feedback.

2 Main Tasks

The design was driven by a single persona scenario: *Narciso moves through remote discovery at home → on-site AR re-emergence → voice dialogue with the avatar → gamified photo hunt*. Modalities were matched to each task's real-world constraints:

- **T1 — Remote Discovery.** Explore a Mapbox map of Navigli access points, compare past/present imagery via a slider, and get directions. *Modalities:* Touch + GPS context.
- **T2 — Photo Game.** Physically locate the vantage point of a historical photograph guided by audio proximity feedback. *Modalities:* GPS + Spatial Audio.
- **T3 — AR Re-emergence.** Place the camera on a street, say “*Bring the water back!*” and watch the canal materialise in AR. *Modalities:* Camera SLAM + Voice command.

- **T4 — Conversational Sciura.** Speak freely with the avatar about canal history in a hands-free, eyes-up loop. *Modalities:* Voice (STT → LLM → TTS).

3 Prototype Architecture

The stack is deliberately decoupled so heavy WebGL rendering and low-latency speech never block the main UI thread.

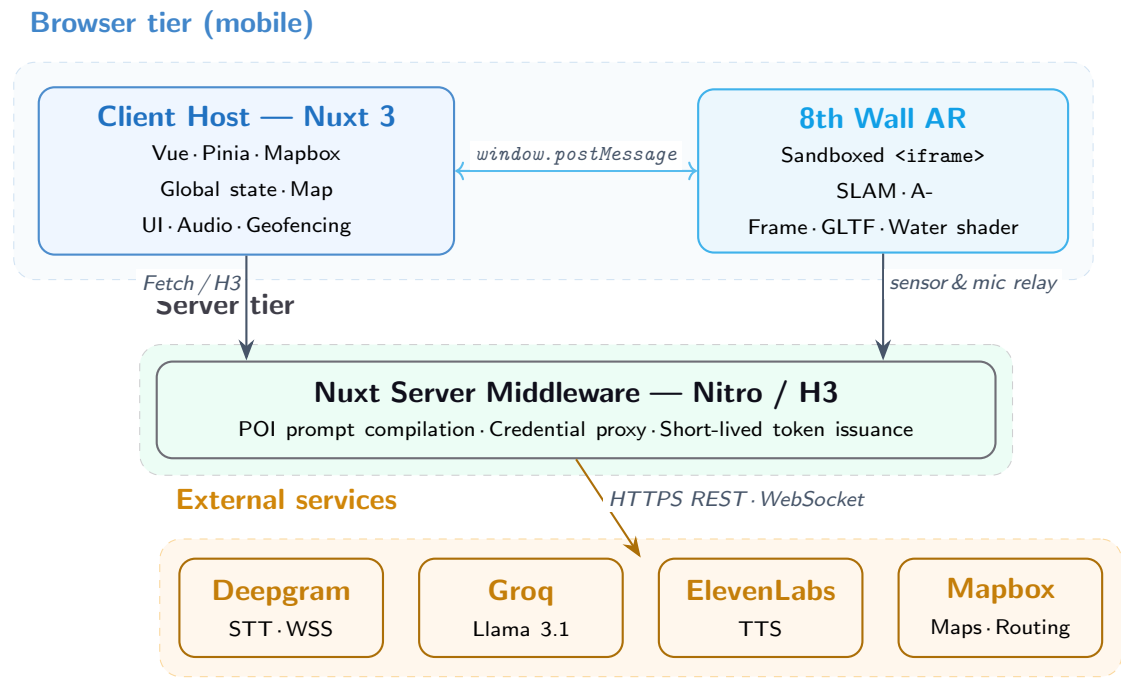


Figure 1: Three-tier architecture of Persephonē. The AR iframe is sandboxed so WebGL/SLAM cannot stall the host UI, map, or audio loops.

Layer Responsibilities			
Layer	Technology	Key responsibilities	Channel
Client Host	Nuxt 3 (Vue, Pinia, Mapbox)	State, map UI, Geiger audio loop, routing	postMessage, Fetch
8th Wall AR	8th Wall / A-Frame	SLAM, GLTF assets, water shader animation	Sandbox iframe, sensors
Server	Nuxt H3 (Nitro API)	Prompt compilation, credential proxy	JSON / H3 events
External	Groq · Deepgram · 11Labs · Mapbox	Real-time STT, LLM persona, TTS, map data	WebSockets, HTTPS REST

3.1 Sequence Diagram — Task 1: Remote Discovery

The remote discovery flow documents the user's interaction with the interactive Navigli map. The user selects a Point of Interest (e.g., Via Senato), triggering a `extttGET /api/pois/{id}` request to the Nuxt server. The server reads the local `pois.json` file, filters the entry, and returns the JSON payload. The client renders a bottom sheet with the past/present slider and directions button.

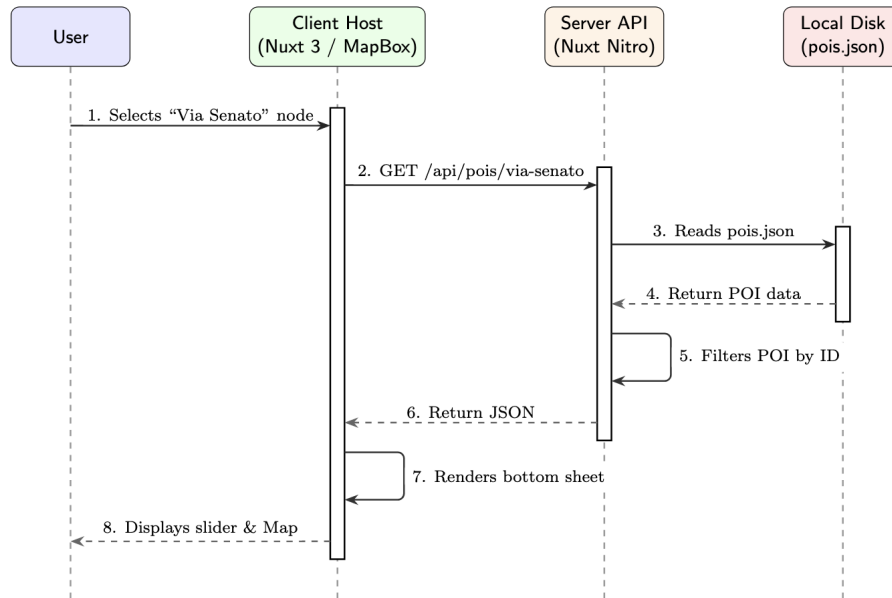


Figure 2: Task 1 — Remote Discovery: map exploration, POI data fetch, and past/present slider.

3.2 Sequence Diagram — Task 2: Photo Game

The Photo Game flow details the real-time GPS geofencing loop. Once the user launches the game and grants location permissions, the tracker component polls the browser’s Geolocation API and computes Haversine distance to the target POI coordinates. This distance drives the Web Audio API tick rate: ticks accelerate as the user approaches the target. On tapping “Check Location” within the geofence, a success chime plays and the historical image gallery is unlocked.

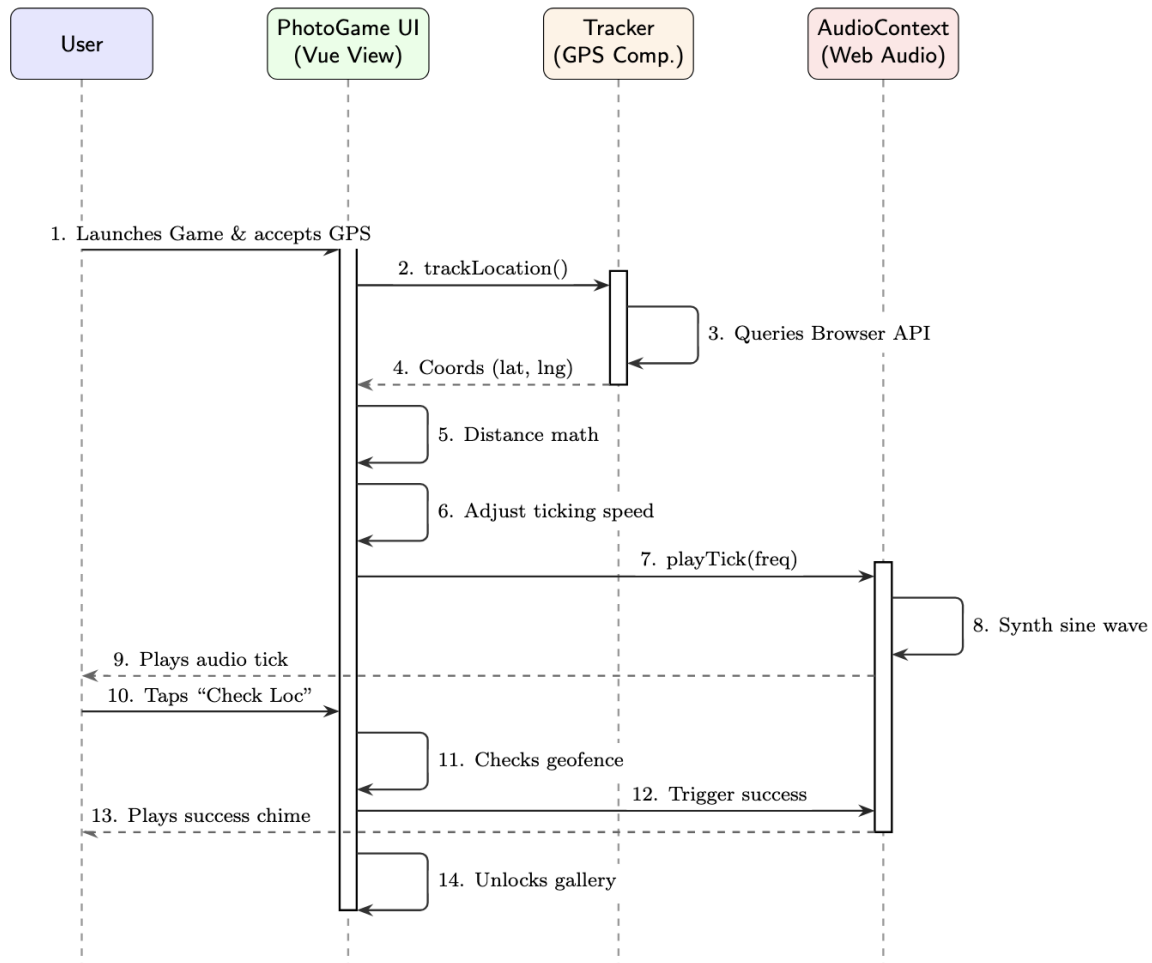


Figure 3: Task 2 — Photo Game: GPS Haversine loop driving Web-Audio Geiger-style acoustic feedback.

3.3 Sequence Diagram — Task 3: AR Re-emergence

The AR Re-emergence flow covers 8th Wall SLAM initialisation inside the sandboxed iframe, Deepgram WebSocket voice capture, and the cross-iframe `TRIGGER_WATER` message. After the user places the 3D canal model by tapping the street, the host page opens the microphone and connects to Deepgram over WSS. When the recognised transcript matches the trigger phrase (“Bring the water back!”), the host posts `TRIGGER_WATER` via `postMessage` to the AR iframe, activating the dynamic water-fill shader.

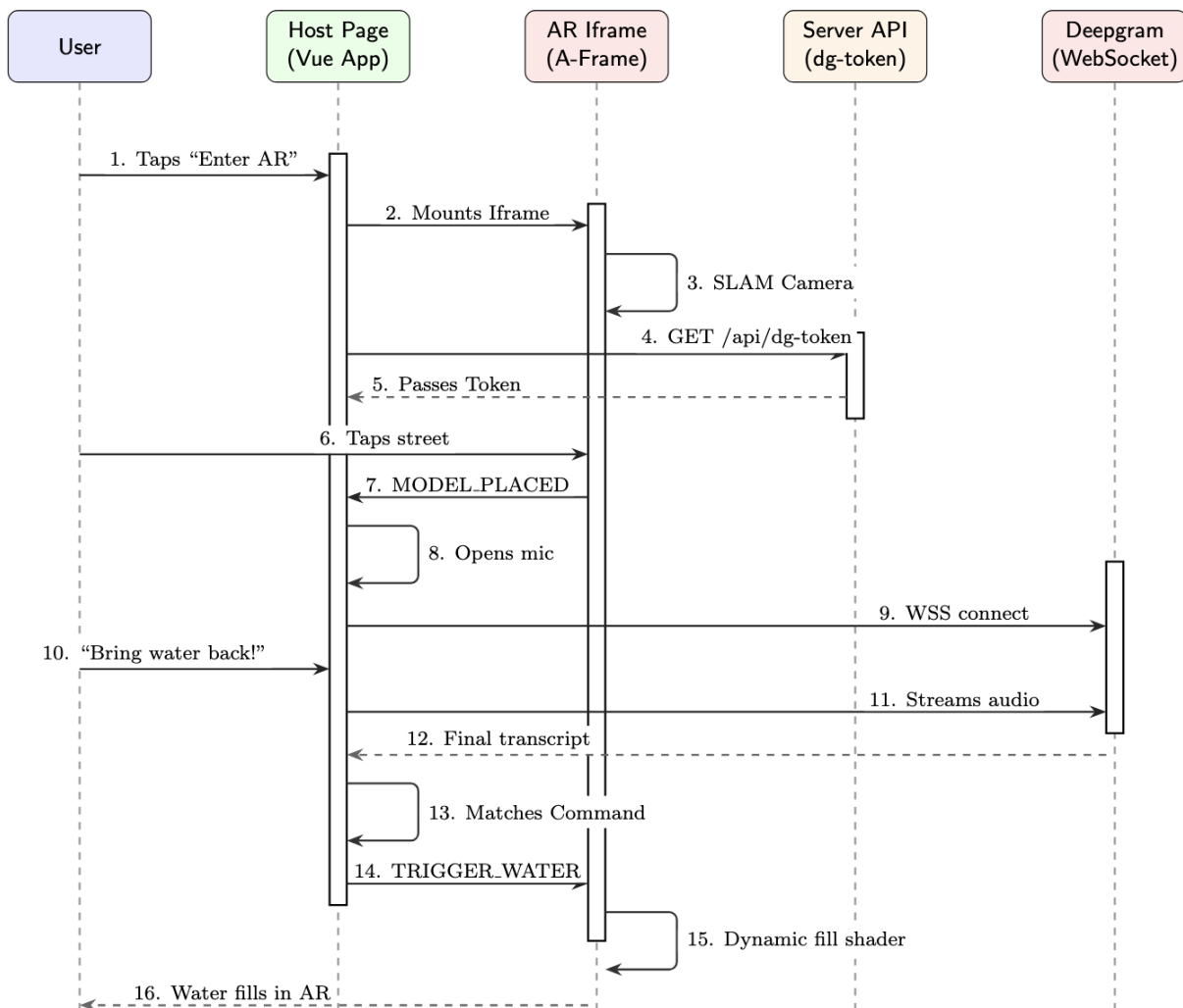


Figure 4: Task 3 — AR Re-emergence: voice command → Deepgram STT → `TRIGGER_WATER` → water shader.

3.4 Sequence Diagram — Task 4: Conversational Sciura

The conversational flow begins when the AR iframe detects the user is within 3m of the virtual canal (USER_NEAR_MODEL), switching the host into Sciura speech mode. The user's spoken question is streamed to Deepgram STT over WebSocket; the transcript is forwarded to `POST /api/chat`, which builds a persona-aware prompt and calls Groq Llama 3.1. The text response is then sent to `POST /api/tts` (ElevenLabs), and the returned MP3 blob is played back with spatial EQ — completing the hands-free dialogue loop.

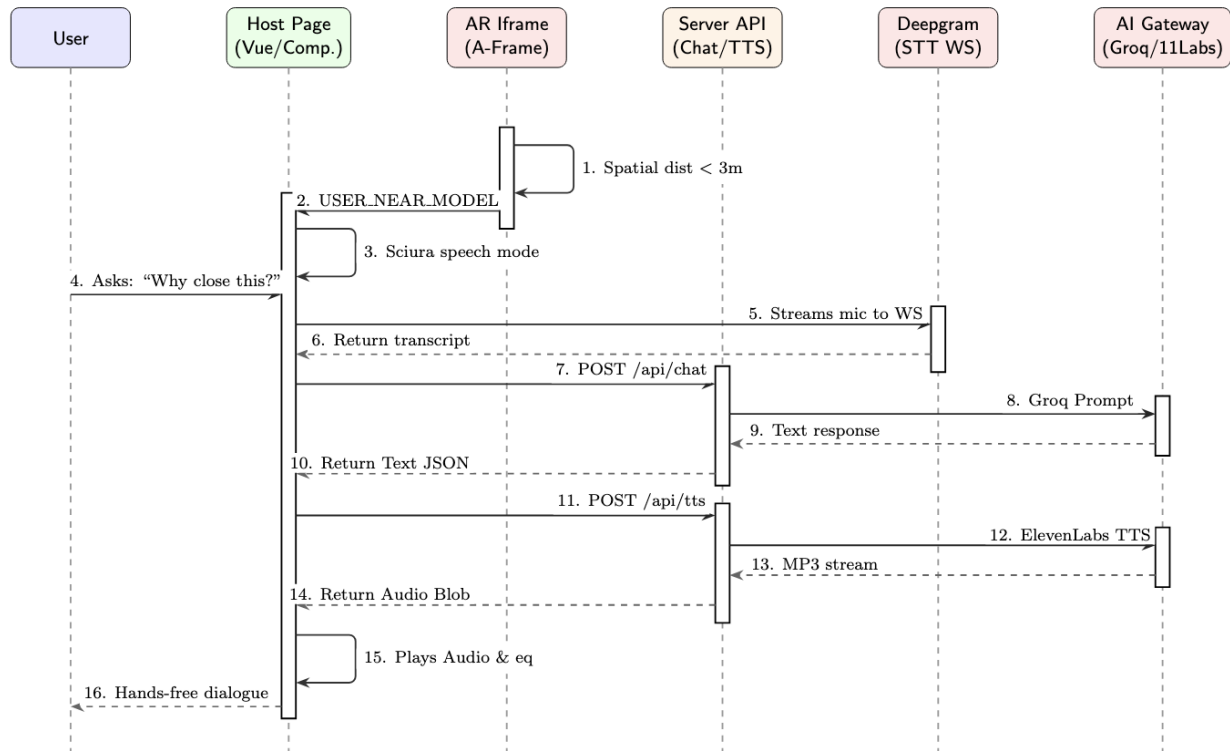


Figure 5: Task 4 — Conversational Sciura: STT → Groq LLM → ElevenLabs TTS → spatial audio.

4 Heuristic Evaluation

Expert-based evaluation against Nielsen’s ten heuristics (H1–H10) adapted for multimodal interaction (modality discoverability, temporal grounding, repair strategies, fallback paths). Tested end-to-end on Android and iOS; **14 findings** across all four tasks.

Key Findings

ID	Finding	Heuristics	Sev.
3.1	Sciura voice fails on Android (permissions/autoplay)	H4, H2	3
2.2	No SLAM scanning feedback during surface detection	H1, H9	3
2.3	No anchoring confirmation; no in-AR re-anchor option	H1, H3, H5	3
2.4	Voice trigger does not echo recognised transcript	H1, H5	3
3.3	User cannot interrupt avatar mid-response	H3	3
3.4	Follow-up questions interpreted as implicit “yes”	H5, H9	3
C.2	No fallback for low-light SLAM failure	H9	3
1.2	“Take me there” silently exits to native Maps	H1, H3	2
2.1	Mic permission before user understands AR flow	H4, H10	2
C.1	GPS banner persists in Italian in English UI	H2, H4, H8	2
1.3	Slider angles diverge for older photographic eras	H2	1

Sev. 3 = Major · 2 = Minor · 1 = Cosmetic. No Sev-4 found.

Confirmed strengths: coherent navigation flow (Map → POI → AR/Game); exemplary text fallback for voice; phrasing-tolerant trigger recognition; in-character network-loss recovery; low conversational latency (~3s).

5 Revisions After Heuristic Evaluation

Eight concrete changes, mapped to findings in two rounds:

Improvements

Finding	Change	What was done
<i>Round 1 — perception & grounding</i>		
1.2	Warn before leaving app	Explicit in-app notice before Maps hand-off; never a silent context switch.
2.3	In-AR re-anchor	Canal can be repositioned without exiting the AR camera view.
2.4 / 3.2	Speech echo	Both voice trigger and Sciura show a written echo of the recognised utterance.
3.1	Android audio fix	Permissions/autoplay root cause identified; resolution in progress with staff.
<i>Round 2 — conversational control & robustness</i>		
3.3	Interruptible Sciura	New input during a response immediately cuts the current playback.
3.4	Remove implicit yes	System prompt rewritten; avatar now respects explicit “no” / topic change.
C.1	Localise GPS banner	Warning follows UI language and can be dismissed.
C.2	Low-light SLAM alert	Failure detected; user given actionable guidance.